

```
#Answer:05-07
def celsius_to_fahrenheit(temp_c):
    temp_f= ((temp_c * 9/5) + 32)
    return(temp_f)
celcius = float(input(" Enter the value: "))
farenhite= celsius_to_fahrenheit(celcius)
print("Temperature in Fahrenheit:", farenhite)

def increase_temperature(f_temp, value=2.0):
    new_t= f_temp+value
    return new_t

default_value= increase_temperature(farenhite)
print("After 2.0 increass", default_value)
user_value= float(input("Enter the value: "))
custome_result= increase_temperature(farenhite, user_value)
print("after user adding value",custome_result )

def display_temperature(name, original_temp, adjusted_temp):
    message="hello "+name+", the temperature is "+ str(original_temp)+"°F+", and adjusted is "+ str(adjusted_temp)
    return message

name=input("Enter your name")
default_result= display_temperature(name, farenhite, default_value)
print(default_result)
coustomer_result= display_temperature(name, farenhite, custome_result)
print(coustomer_result)
```

```
Enter the value: 25
Temperature in Fahrenheit: 77.0
After 2.0 increass 79.0
Enter the value: 5
after user adding value 82.0
Enter your nameSabir
hello Sabir, the temperature is 77.0°F, and adjusted is 79.0°F
hello Sabir, the temperature is 77.0°F, and adjusted is 82.0°F
```

```
def gcd (a,b):
    if b==0:
        return a
    else:
        return gcd(b,a%b)
def lcm (a,b):
    return (a*b)//gcd (a,b)
a= int(input("Enter the firs value: "))
b= int(input("Enter the second value: "))
print("GCD=",gcd (a,b))
print("LCM=",lcm(a,b))
```

```
Enter the firs value: 43
Enter the second value: 66
GCD= 1
LCM= 2838
```

```
#Answer:01-02
def greet_user(name):
    if name == "":
        name= "Guest"
    message= "welcome,"+ name
    return message
name= input("Enter your name: ")
user_message= greet_user(name)
print(user_message)
```

```
enter your name: Sabir
welcome,Sabir
```

```
#Answer:03
def square(num):
    result= num*num
    return result
number= float(input("Enter your valu: "))
user_value= square(number)
print(user_value)
```

Enter your valu: 25.5
650.25

```
#Answer:04
def introduce(name, age):
    message = name+" is "+ str(age) +" years old."
    return message
name= input("Enter the name: ")
age= int(input("Enter your age: "))
user_message= introduce(name, age)
print(user_message)
```

Enter the name: sabir
Enter your age: 66
sabir is 66 years old.

```
#Answer:04 or
def introduce(name, age):
    print(f"{name} is {age} years old. ")
introduce(name="sabir", age=22)
```

sabir is 22 years old.

```
#Answer:08
def check_adult(age):

    if age>=18:
        print ("You are an adult.")
    else:
        print("You are a minor.")

age= int(input("Enter your age: "))
check_adult(age)
```

Enter your age: 9
You are a minor.

```
#Answer:08 or
def check_adult(age):

    if age>=18:
        return ("You are an adult.")
    else:
        return ("You are a minor.")

age= int(input("Enter your age: "))
print(check_adult(age))
```

Enter your age: 8
You are a minor.

```
#Answer:09
def grade(score):
    if score >=90:
        return ("Grade A ")
    elif score >=80:
        return ("Grade B")
    else:
        return ("Grade C or below")

score= float(input("Enter your mark: "))
print(grade(score))
```

Enter your mark: 90
Grade A

```
#Answr:10
def check_range(num):
    if 10<= num <=20:
        return ("whether a number is in the range 10 to 20 (inclusive)")
    else:
        return("whether a number is not in the range 10 to 20 (inclusive)")
num= int(input("Enter your value: "))
print(check_range(num))
```

Enter your value: 20
whether a number is in the range 10 to 20 (inclusive)

```
#Answer:11
def get_discount_message(age):
    if age< 12 or age>60:
        return ("You qualify for a discount.")
    else:
        return("No discount available.")

age= int(input("Enter your age: "))
print(get_discount_message(age))
```

Enter your age: 30
No discount available.

```
#Answer:12
def my_funcation():
    x=5
    return x
x=my_funcation()
print("outside funcation:",x)
```

outside funcation: 5

```
#Answer:13
def update(num):
    num=num+10
    return num
num=int(input("Enter the value: "))
update_num=update(num)
print("update value:" ,update_num)
```

Enter the value: 5
update value: 15

```
#Answer:14
def compare(a, b):
    if a==b:
        return("Equal")
    elif a>b:
        return ("A is greater")
    else:
        return ("B is greater")
a=int(input("Enter the value of A: "))
b=int(input("Enter the value of B: "))
compare_num= compare(a, b)
print("the compare of number is:", compare_num)
```

Enter the value of A: 12
Enter the value of B: 3
the compare of number is: A is greater

```
#Answer:15
def weather_report(name, temp_c):
    temp_f=((temp_c * 9/5) + 32)
    return temp_f
name=input("enter your name: ")
temp_c=int(input("Enter your celcius value: "))
farenhite= weather_report(name, temp_c)
print(farenhite)

def custom_offset(temp_f , default=2.00):
    temp_f= temp_f+default
    return temp_f

default_value= custom_offset(farenhite)
print(default_value)

def display_temperature(name, temp_f,coustom_f):
    message= name + " the temperature is "+ str(temp_f)+"°F"+", and coustom is "+ str(coustom_f)+"°F"
    return message

coustom_message= display_temperature(name, farenhite, default_value)
print(coustom_message)
```

enter your name: sabir
Enter your celcius value: 25
77.0
79.0
sabir the temperature is 77.0°F, and coustom is 79.0°F

```
#Answer:16
def compare_temperatures(city1, temp1, city2, temp2):
    if temp1>temp2:
        return f"{city1} is hotter than {city2}."
    elif temp1<temp2:
        return f"{city2} is hotter than {city1}."
    else:
        return f"{city1} and {city2} temperature is equal"

city1=input("Enter the city1: ")
city2=input("Enter the city2: ")
temp1=int(input("Enter your temp1: "))
temp2=int(input("Enter your temp2: "))
temperature_message= compare_temperatures(city1, temp1, city2, temp2)
print(temperature_message)
```

```
Enter the city1: magura
Enter the city2: dhaka
Enter your temp1: 15
Enter your temp2: 25
dhaka is hotter than magura.
```

```
#Answer:17
def calculate_bill(amount, discount=0):
    return amount - discount
amount= float(input("Enter your amount: "))
discount= float(input("Enter discount amount: "))
bill_message= calculate_bill(amount,discount)
print("final bill:",bill_message)
```

```
Enter your amount: 1500
Enter discount amount: 300
final bill: 1200.0
```

```
#Answer:18
def check_login(username, password):
    if username == "admin" and password == "1234":
        return "Access granted"
    else:
        return "Access denied"
username= input("enter your username: ")
password= input("enter your password: ")
check_message=check_login(username, password)
print(check_message)
```

```
enter your username: admin
enter your password: 1234
Access granted
```

```
#Answer:19
def describe_person(name, age, country="Bangladesh"):
    message= " Hello, " + name + " is " + str(age) + " years old from " + country
    return message
name=input("enter your name: ")
age=int(input("Enter your age: "))
describe_message=describe_person(name, age, country="Bangladesh")
print(describe_message)
```

```
enter your name: Sabir
Enter your age: 23
Hello, Sabir is 23 years old from Bangladesh
```

```
#Answer:20
def max_of_three(a, b, c):
    if a>=b and a>=c:
        return a
    elif b>=a and b>=c:
        return b
    else:
        return c

a=int(input("enter your a value : "))
b=int(input("enter your b value : "))
c=int(input("enter your c value : "))
result= max_of_three(a, b, c)
print (result)
```

```
enter your a value : 12
enter your b value : 265
```

```
enter your c value : 23
265
```

```
#Answ:21
def check_eligibility(age, has_id):
    if age>=18 and has_id == True:
        return "Eligible"
    else:
        return "Not Eligible"
age=int(input("Enter your age: "))
has_id= input("Enter your id: ")=="True"
check=check_eligibility(age, has_id)
print(check)
```

```
Enter your age: 20
Enter your id: True
Eligible
```

```
#Answer:22
def convert_and_adjust(temp_c, offset=2.5):
    temp_f=((temp_c * 9/5) + 32)
    adjusted_fahrenheit = temp_f+ offset
    message= "The orginal tampurate is "+ str(temp_f)+("°F")+", and the adjust tampurate is "+ str(adjusted_fahren
    return (message)
temp_c= int(input("Enter your tempurature: "))
check= convert_and_adjust(temp_c, offset=2.5)
print (check)
```

```
Enter your temperature: 20
The orginal tampurate is 68.0°F, and the adjust tampurate is 70.5°F
```

```
#Answr:22 another
def convert_and_adjust(temp_c, offset=2.5):
    temp_f=((temp_c * 9/5) + 32)
    adjusted_fahrenheit = temp_f+ offset
    return (temp_f,adjusted_fahrenheit)
temp_c= int(input("Enter your temperature: "))
check= convert_and_adjust(temp_c, offset=2.5)
print (check)
```

```
Enter your temperature: 20
(68.0, 70.5)
```

```
#Answer:23
def check_number(num):
    if num ==0:
        return "Zero"
    elif num % 2 ==0:
        return "Even"
    else:
        return "odd"

num= int(input("Enter your number: "))
print(check_number(num))
```

```
Enter your number: 0
Zero
```

```
#Answer:24
def inside_vs_outside():
    x = 5
    print("Inside function:", x)
inside_vs_outside()
print("Outside function:", x)
```

```
Outside function: 5
```

```
#Answer:25
def update_string(text):
    text= "update_string"
    print("inside funcation:",text)

my_text= "Hello"
print("Before funcaton call:",my_text)
update_string(my_text)
print("After funcation call:",my_text)
```

```
Before funcaton call: Hello
inside funcation: update_string
After funcation call: Hello
```

```
#Answer:26
def apply_discount(price, percent=10):
    discount=((price*percent)/100)
    final_price=(price-discount)
    return (discount,final_price)
price= float(input("Enter the total Amount: "))
discount,final_price = apply_discount(price)
print("The Discount Amount is: ",discount)
print("The Final Amount After Discount: ",final_price)
```

Enter the total Amount: 200
 The Discount Amount is: 20.0
 The Final Amount After Discount: 180.0

```
#Answer:27
def multi_check(num1, num2):
    if num1 % 3 == 0 and num2 % 3 == 0:
        return "Both numbers are divisible by 3"
    elif num1 % 2 == 0 or num2 % 2 == 0:
        return "At least one number is even"
    else:
        return "Neither condition is satisfied"
num1 = int(input("Enter your number1: "))
num2 = int(input("Enter your number2: "))
print(multi_check(num1, num2))
```

Enter your number1: 20
 Enter your number2: 30
 At least one number is even

```
#Answer:28
def secret_message(name, code):
    if code == "open":
        msg = "Welcome, " + name + "!"
    else:
        msg = "Access Blocked"
    return msg
name= input("Enter your name: ")
code= input("enter the code: ")
print(secret_message(name,code))
```

Enter your name: sabir
 enter the code: open
 Welcome, sabir!

```
#Answer:29
def temperature_advice(temp_f):
    if temp_f > 100:
        return "Too hot"
    elif temp_f < 32:
        return "Too cold"
    else:
        return "Moderate"
temp_f=float(input("Enter your temprature: "))
print(temperature_advice(temp_f))
```

Enter your temprature: 109.90
 Too hot

```
#Answer:30
def weather_report(name, temp_c, offset=2):
    temp_f=((temp_c * 9/5) + 32)
    Adds_offset= temp_f+offset
    if Adds_offset > 100:
        advice = "Stay hydrated!"
    elif Adds_offset < 50:
        advice = "Wear a jacket!"
    else:
        advice = "Enjoy the weather!"
    message= "Hello " + name + "! " + advice + " Current temperature is " + str(Adds_offset) + "°F."
    return message
name=input("Enter your name: ")
temp_c= float(input("Enter your temprature: "))
print(weather_report(name, temp_c))
```

Enter your name: sabir
 Enter your temprature: 700
 Hello sabir! Stay hydrated! Current temperature is 1294.0°F.